

2

Probability

Contents

2 Probability

Introduction

2.1 Sample Spaces and Events

2.2 Axioms, Interpretations,
and Properties of Probability

2.3 Counting Techniques

2.4 Conditional Probability

2.5 Independence

2.3 Counting Techniques

Counting Techniques

When the various outcomes of an experiment are equally likely (the same probability is assigned to each simple event), the task of computing probabilities reduces to counting.

Letting N denote the number of outcomes in a sample space and $N(A)$ represent the number of outcomes contained in an event A ,

$$P(A) = \frac{N(A)}{N} \quad (2.1)$$

Counting Techniques

If a list of the outcomes is easily obtained and N is small, then N and $N(A)$ can be determined without the benefit of any general counting principles.

There are, however, many experiments for which the effort involved in constructing such a list is prohibitive because N is quite large.

By exploiting some general counting rules, it is possible to compute probabilities of the form (2.1) without a listing of outcomes.

These rules are also useful in many problems involving outcomes that are not equally likely.



The Product Rule for Ordered Pairs

github.com/saeedgrad

The Product Rule for Ordered Pairs

Our first counting rule applies to any situation in which a set (event) consists of ordered pairs of objects and we wish to count the number of such pairs.

By an ordered pair, we mean that, if O_1 and O_2 are objects, then the pair (O_1, O_2) is different from the pair (O_2, O_1) .

For example, if an individual selects one airline for a trip from Los Angeles to Chicago and (after transacting business in Chicago) a second one for continuing on to New York, one possibility is (American, United), another is (United, American), and still another is (United, United).

The Product Rule for Ordered Pairs

Proposition

If the first element or object of an ordered pair can be selected in n_1 ways, and for each of these n_1 ways the second element of the pair can be selected in n_2 ways, then the number of pairs is $n_1 n_2$.

If the first element or object of an ordered pair can be selected in n_1 ways, and for each of these n_1 ways the second element of the pair can be selected in n_2 ways, then the number of pairs is $n_1 n_2$.

github.com

The Product Rule for Ordered Pairs

An alternative interpretation involves carrying out an operation that consists of two stages.

If the first stage can be performed in any one of n_1 ways, and for each such way there are n_2 ways to perform the second stage, then $n_1 n_2$ is the number of ways of carrying out the two stages in sequence.

github.com/saeedgrad

Example 18

A family has just moved to a new city and requires the services of both an obstetrician and a pediatrician. There are two easily accessible medical clinics, each having two obstetricians and three pediatricians.

The family will obtain maximum health insurance benefits by joining a clinic and selecting both doctors from that clinic. In how many ways can this be done?

Denote the obstetricians by O_1, O_2, O_3 , and O_4 and the pediatricians by $P_1, \dots, P_6$.

Example 18

cont'd

Then we wish the number of pairs (O_i, P_j) for which O_i and P_j are associated with the same clinic.

Because there are four obstetricians, $n_1 = 4$, and for each there are three choices of pediatrician, so $n_2 = 3$.

Applying the product rule gives $N = n_1 n_2 = 12$ possible choices.

github.com/saeedgrad

The Product Rule for Ordered Pairs

In many counting and probability problems, a configuration called a **tree diagram** can be used to represent pictorially all the possibilities.

The tree diagram associated with Example 18 appears in Figure 2.7.

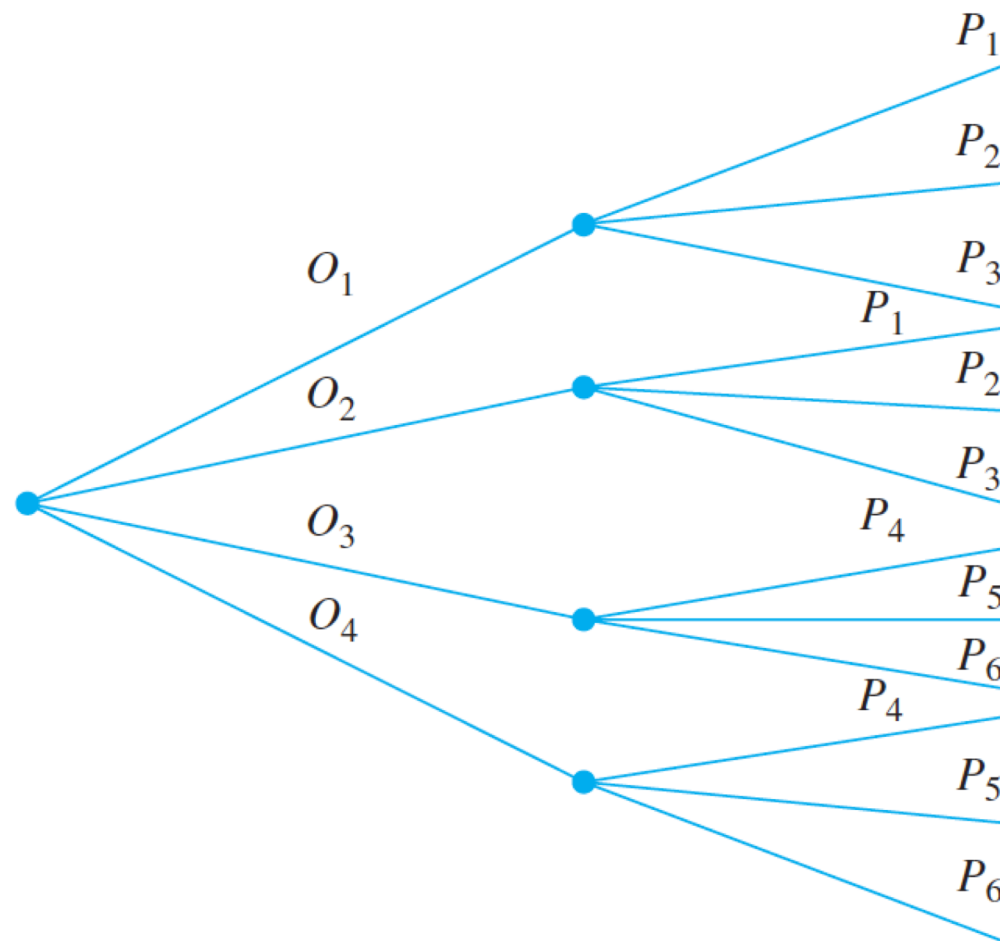


Figure 2.7 Tree diagram for Example 2.18

The Product Rule for Ordered Pairs

Starting from a point on the left side of the diagram, for each possible first element of a pair a straight-line segment emanates rightward.

Each of these lines is referred to as a first-generation branch.

Now for any given first-generation branch we construct another line segment emanating from the tip of the branch for each possible choice of a second element of the pair.

The Product Rule for Ordered Pairs

Each such line segment is a second-generation branch. Because there are four obstetricians, there are four first-generation branches, and three pediatricians for each obstetrician yields three second-generation branches emanating from each first-generation branch.

Generalizing, suppose there are n_1 first-generation branches, and for each first generation branch there are n_2 second-generation branches.

The total number of second-generation branches is then $n_1 n_2$.

The Product Rule for Ordered Pairs

Since the end of each second-generation branch corresponds to exactly one possible pair (choosing a first element and then a second puts us at the end of exactly one second-generation branch), there are $n_1 n_2$ pairs, verifying the product rule.

The construction of a tree diagram does not depend on having the same number of second-generation branches emanating from each first-generation branch.

github.com/saeedgrah

The Product Rule for Ordered Pairs

If the second clinic had four pediatricians, then there would be only three branches emanating from two of the first-generation branches and four emanating from each of the other two first-generation branches.

A tree diagram can thus be used to represent pictorially experiments other than those to which the product rule applies.

github.com/saeedgrad



A More General Product Rule

github.com/saeedgrad

A More General Product Rule

If a six-sided die is tossed five times in succession rather than just twice, then each possible outcome is an ordered collection of five numbers such as $(1, 3, 1, 2, 4)$ or $(6, 5, 2, 2, 2)$.

We will call an **ordered collection of k objects a k -tuple** (so a pair is a 2-tuple and a triple is a 3-tuple).

Each outcome of the die-tossing experiment is then a 5-tuple.

github.com/smedgrad

A More General Product Rule

Product Rule for k -Tuples

Suppose a set consists of ordered collections of k elements (k -tuples) and that there are n_1 possible choices for the first element; for each choice of the first element, there are n_2 possible choices of the second element; . . . ; for each possible choice of the first $k - 1$ elements, there are n_k choices of the k th element. Then there are $n_1 n_2 \cdot \cdot \cdot \cdot n_k$ possible k -tuples.

Then there are $n_1 n_2 \cdot \cdot \cdot \cdot n_k$ possible k -tuples.

A More General Product Rule

An alternative interpretation involves carrying out an operation in k stages.

If the first stage can be performed in any one of n_1 ways, and for each such way there are n_2 ways to perform the second stage, and for each way of performing the first two stages there are n_3 ways to perform the 3rd stage, and so on, then $n_1 n_2 \cdot \cdot \cdot n_k$ is the number of ways to carry out the entire k -stage operation in sequence.

This more general rule can also be visualized with a tree diagram. For the case $k = 3$, simply add an appropriate number of 3rd generation branches to the tip of each 2nd generation branch.

A More General Product Rule

If, for example, a college town has four pizza places, a theater complex with six screens, and three places to go dancing, then there would be four 1st generation branches, six 2nd generation branches emanating from the tip of each 1st generation branch, and three 3rd generation branches leading off each 2nd generation branch.

Each possible 3-tuple corresponds to the tip of a 3rd generation branch.

github.com/saeedgrad

Example 19

Example 17 continued...

Suppose the home remodeling job involves first purchasing several kitchen appliances. They will all be purchased from the same dealer, and there are five dealers in the area.

With the dealers denoted by $D_1, \dots, D_5$, there are $N = n_1 n_2 n_3 = (5)(12)(9) = 540$ 3-tuples of the form (D_i, P_j, Q_k) , so there are 540 ways to choose first an appliance dealer, then a plumbing contractor, and finally an electrical contractor.



Permutations and Combinations

github.com/saeedgrad

Permutations and Combinations

Consider a group of n distinct individuals or objects (“distinct” means that there is some characteristic that differentiates any particular individual or object from any other).

How many ways are there to select a subset of size k from the group?

For example, if a Little League team has 15 players on its roster, how many ways are there to select 9 players to form a starting lineup?

Permutations and Combinations

Or if a university bookstore sells ten different laptop computers but has room to display only three of them, in how many ways can the three be chosen?

An answer to the general question just posed requires that we distinguish between two cases. In some situations, such as the baseball scenario, the order of selection is important.

For example, Angela being the pitcher and Ben the catcher gives a different lineup from the one in which Angela is catcher and Ben is pitcher.

Permutations and Combinations

Often, though, order is not important and one is interested only in which individuals or objects are selected, as would be the case in the laptop display scenario.

Definition

An **ordered subset** is called a **permutation**. The number of permutations of size k that can be formed from the n individuals or objects in a group will be denoted by $P_{k,n}$.

Definition

An **unordered subset** is called a **combination**. One way to denote the number of combinations is $C_{k,n}$, but we shall instead use notation that is quite common in probability books: $\binom{n}{k}$, read “ n choose k ”.

Permutations and Combinations

The number of permutations can be determined by using our earlier counting rule for k -tuples. Suppose, for example, that a college of engineering has seven departments, which we denote by a, b, c, d, e, f , and g .

Each department has one representative on the college's student council. From these seven representatives, one is to be chosen chair, another is to be selected vice-chair, and a third will be secretary.

How many ways are there to select the three officers? That is, how many permutations of size 3 can be formed from the 7 representatives?

Permutations and Combinations

To answer this question, think of forming a triple (3-tuple) in which the first element is the chair, the second is the vice-chair, and the third is the secretary.

One such triple is (a, g, b) , another is (b, g, a) , and yet another is (d, f, b) . Now the chair can be selected in any of $n_1 = 7$ ways.

For each way of selecting the chair, there are $n_2 = 6$ ways to select the vice-chair, and hence $7 \times 6 = 42$ (chair, vice-chair) pairs.

Permutations and Combinations

Finally, for each way of selecting a chair and vice-chair, there are $n_3 = 5$ ways of choosing the secretary. This gives

$$P_{3,7} = (7)(6)(5) = 210$$

as the number of permutations of size 3 that can be formed from 7 distinct individuals. A tree diagram representation would show three generations of branches.

The expression for $P_{3,7}$ can be rewritten with the aid of *factorial notation*. Recall that $7!$ (read “7 factorial”) is **compact notation** for the descending product of integers $(7)(6)(5)(4)(3)(2)(1)$.

Permutations and Combinations

More generally, for any positive integer m ,

$m! = m(m - 1)(m - 2) \cdot \cdot \cdot \cdot (2)(1)$ This gives $1! = 1$, and we also define $0! = 1$. Then

$$P_{3,7} = (7)(6)(5) = \frac{(7)(6)(5)(4!)}{(4!)} = \frac{7!}{4!}$$

More generally,

$$P_{k,n} = n(n - 1)(n - 2) \cdot \cdot \cdot \cdot (n - (k - 2))(n - (k - 1))$$

Multiplying and dividing this by $(n - k)!$ gives a compact expression for the number of permutations.

Permutations and Combinations

Proposition

$$P_{k,n} = \frac{n!}{(n-k)!}$$

github.com/saeedgrad

Example 21

There are ten teaching assistants available for grading papers in a calculus course at a large university.

The first exam consists of four questions, and the professor wishes to select a different assistant to grade each question (only one assistant per question).

In how many ways can the assistants be chosen for grading? Here n = group size = 10 and k = subset size = 4.

github.com/saeedghad

Example 21

cont'd

The number of permutations is

$$P_{4,10} = \frac{10!}{(10 - 4)!} = \frac{10!}{6!} = 10(9)(8)(7) = 5040$$

That is, the professor could give 5040 different four-question exams without using the same assignment of graders to questions, by which time all the teaching assistants would hopefully have finished their degree programs!

github.com/saeedghad

Permutations and Combinations

Now let's move on to combinations (i.e., unordered subsets).

Again refer to the student council scenario, and suppose that three of the seven representatives are to be selected to attend a statewide convention.

The order of selection is not important; all that matters is which three get selected. So we are looking for $\binom{7}{3}$, the number of combinations of size 3 that can be formed from the 7 individuals.

Permutations and Combinations

Consider for a moment the combination a, c, g .

These three individuals can be ordered in $3! = 6$ ways to produce permutations:

a, c, g a, g, c c, a, g c, g, a g, a, c g, c, a

Similarly, there are $3! = 6$ ways to order the combination b, c, e to produce permutations, and in fact $3!$ ways to order any particular combination of size 3 to produce permutations.

Permutations and Combinations

This implies the following relationship between the number of combinations and the number of permutations:

$$P_{3,7} = (3!) \cdot \binom{7}{3} \Rightarrow \binom{7}{3} = \frac{P_{3,7}}{3!} = \frac{7!}{(3!)(4!)} = \frac{(7)(6)(5)}{(3)(2)(1)} = 35$$

It would not be too difficult to list the 35 combinations, but there is no need to do so if we are interested only in how many there are.

Notice that the number of permutations 210 far exceeds the number of combinations; the former is larger than the latter by a factor of 3! since that is how many ways each combination can be ordered.

Permutations and Combinations

Generalizing the foregoing line of reasoning gives a simple relationship between the number of permutations and the number of combinations that yields a concise expression for the latter quantity.

Proposition

$$\binom{n}{k} = \frac{P_{k,n}}{k!} = \frac{n!}{k!(n-k)!}$$

Notice that $\binom{n}{n} = 1$ and $\binom{n}{0} = 1$ since there is only one way to choose a set of (all) n elements or of no elements, and $\binom{n}{1} = n$ since there are n subsets of size 1.

Example 22

A particular iPod playlist contains 100 songs, 10 of which are by the Beatles.

Suppose the shuffle feature is used to play the songs in random order (the randomness of the shuffling process is investigated in “Does Your iPod *Really* Play Favorites?”

What is the probability that the first Beatles song heard is the fifth song played?

In order for this event to occur, it must be the case that the first four songs played are not Beatles' songs (NBs) and that the fifth song is by the Beatles (B).

Example 22

cont'd

The number of ways to select the first five songs is $100(99)(98)(97)(96)$.

The number of ways to select these five songs so that the first four are NBs and the next is a B is $90(89)(88)(87)(10)$.

The random shuffle assumption implies that any particular set of 5 songs from amongst the 100 has the same chance of being selected as the first five played as does any other set of five songs; each outcome is equally likely.

Example 22

cont'd

Therefore the desired probability is the ratio of the number of outcomes for which the event of interest occurs to the number of possible outcomes:

$$P(1^{\text{st}} \text{ B is the } 5^{\text{th}} \text{ song played}) = \frac{90 \cdot 89 \cdot 88 \cdot 87 \cdot 10}{100 \cdot 99 \cdot 98 \cdot 97 \cdot 96} = \frac{P_{4,90} \cdot (10)}{P_{5,100}} = .0679$$

Here is an alternative line of reasoning involving combinations.

Rather than focusing on selecting just the first five songs, think of playing all 100 songs in random order.

Example 22

cont'd

The number of ways of choosing 10 of these songs to be the Bs (without regard to the order in which they are then played) is $\binom{100}{10}$.

Now if we choose 9 of the last 95 songs to be Bs, which can be done in $\binom{95}{9}$ ways, that leaves four NBs and one B for the first five songs.

There is only one further way for these five to start with four NBs and then follow with a B (remember that we are considering *unordered* subsets).

Example 22

cont'd

Thus

$$P(1^{\text{st}} \text{ B is the } 5^{\text{th}} \text{ song played}) = \frac{\binom{95}{9}}{\binom{100}{10}}$$

It is easily verified that this latter expression is in fact identical to the first expression for the desired probability, so the numerical result is again .0679.

github.com/saeedgrad

Example 22

cont'd

The probability that one of the first five songs played is a Beatles' song is

$P(1^{\text{st}} \text{ B is the } 1^{\text{st}} \text{ or } 2^{\text{nd}} \text{ or } 3^{\text{rd}} \text{ or } 4^{\text{th}} \text{ or } 5^{\text{th}} \text{ song played})$

$$= \frac{\binom{99}{9}}{\binom{100}{10}} + \frac{\binom{98}{9}}{\binom{100}{10}} + \frac{\binom{97}{9}}{\binom{100}{10}} + \frac{\binom{96}{9}}{\binom{100}{10}} + \frac{\binom{95}{9}}{\binom{100}{10}} = .4162$$

It is thus rather likely that a Beatles' song will be one of the first five songs played. Such a “coincidence” is not as surprising as might first appear to be the case.